

- Contingency planning for longer than expected queue lengths should be included as part of the TGS design.
- End of queue protection measures must be provided in accordance with Section 4.8.3.

Table 5.4 – Recommended maximum length of operation under shuttle flow

Traffic volume (Total for both directions) (vph)	Length of single lane section (m)
701–800	70
601–700	100
501–600	150
401–500	250
351–400	400
301–350	600
≤ 300	800

The volumes in Table 5.4 have been determined to allow a quick analysis without referring to a traffic engineering professional. Additional traffic engineering input is required to support a longer length of single-lane operation.

Generally, when using Table 5.4, and where the lengths are within the maximum limits, single-lane operation using active control by portable traffic control devices or traffic controllers will lead to a relatively short and consistent or stable queue length; however, additional traffic engineering input and considerations (risk assessments) are required to support longer lengths of single-lane operation which will generally lead to longer maximum queue lengths and queue lengths that are not easily managed, or are variable and unstable.

Figure 5.4.4(a) does not include all traffic control devices required and is not to be used as a TGS diagram.